

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1516

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 15
Duration: 60 Minutes

Total Marks: 30

Code of Conduct

I __Y SRAVAN KUMAR (192472289) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____SRAVAN_____

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

Assessment Tasks

MCQ Test

- 1. A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?**
 - a) 5 TB
 - b) 10 TB
 - c) 15 TB
 - d) 20 TB
- 2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?**
 - a) ~4 minutes
 - b) ~40 minutes
 - c) ~4 hours
 - d) ~24 hours
- 3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?**
 - a) 5000
 - b) 6000
 - c) 7000
 - d) 10,000
- 4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?**
 - a) 1 hour
 - b) 10 hours
 - c) 1.1 hours
 - d) 5.5 hours
- 5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?**
 - a) ~4 hours
 - b) ~8 hours
 - c) ~20 hours
 - d) ~40 hours
- 6. A cloud provider offers pay-as-you-go pricing at \$0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?**

- a) \$50
- b) \$500
- c) \$1000
- d) \$2000

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

- a) 9.8 Gbps
- b) 9.6 Gbps
- c) 8.0 Gbps
- d) 7.5 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

- a) 91.25 GB
- b) 85 GB
- c) 100 GB
- d) 120 GB

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

- a) 500 Mbps
- b) 250 Mbps
- c) 200 Mbps
- d) 1000 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

- a) 262 minutes
- b) 365 minutes
- c) 525 minutes
- d) 150 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

- a) 4,000
- b) 8,000
- c) 6,000
- d) 10,000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?
- a. 4 GB
 - b) 6 GB
 - c) -8 GB
 - d) 10 GB
13. **A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?**
- a. 16
 - b) 32
 - c) 64
 - d) 128
14. **A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?**
- a. 40 kW
 - b) 50 kW
 - c) 60 kW
 - d) 70 kW
15. **A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?**
- a. 6,400
 - b) 12,800
 - c) 25,600
 - d) 50,000
16. Which cloud deployment model is used for organizations that require both private and public cloud resources?
- A): Hybrid Cloud
 - B): Public Cloud
 - C): Private Cloud
 - D): Community Cloud

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

- A): Encryption
- B): Load balancing
- C): Virtualization
- D): Resource pooling

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

- a:** 5000 GB
- b:** 7000 GB
- c:** 6000 GB
- d:** 7500 GB

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

- a:** 56 GHz
- b:** 58 GHz
- c:** 64 GHz
- d:** 70 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

- a:** 1.5 sec
- b:** 2 sec
- c:** 1 sec
- d:** 3 sec

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

- a:** 600 GB
- b:** 500 GB
- c:** 700 GB
- d:** 800 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

- a:** 21.6 min
- b:** 36 min
- c:** 43.2 min

d: 18 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

a: 120 MB

b: 100 MB

c: 150 MB

d: 200 MB

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.

a: 180

b: 150

c: 120

d: 160

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

a: 1200

b: 1000

c: 800

d: 600

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

a: 12 min

b: 10 min

c: 6 min

d: 8 min

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.

Formula:

Energy = Units $\times$ Power $\times$ Time (hrs)

a: 420 kWh

b: 480 kWh

- c:** 400 kWh
- d:** 500 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

- a:** 32 TB
- b:** 35 TB
- c:** 30 TB
- d:** 28 TB

29. Which of the following is NOT a benefit of virtualization?

- i. Reduced hardware costs
- b) Increased energy consumption
- c) Improved resource management
- d) Faster deployment of applications

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

- i. Hypervisor
- b) Load balancer
- c) Firewall
- d) BIOS

MCQ Test

1. A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?

Answer: c) 15 TB

Explanation: $5 \text{ TB} \times 3 \text{ regions} = 15 \text{ TB total storage}$.

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

Answer: a) ~4 minutes

Explanation: 0.01% of a 30-day month is approximately 4.32 minutes.

3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

Answer: b) 6000

Explanation: $60,000 \div 10 = 6,000 \text{ requests/second/server}$.

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

Answer: c) 1.1 hours

Explanation: $500 \text{ GB} \times 8 = 4,000 \text{ Gb}$; $4,000 \div 1 \text{ Gbps} = 4,000 \text{ seconds} \approx 1.1 \text{ hours}$.

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

Answer: b) ~4.4 hours

Explanation: 0.05% of 365 days is approximately 4.38 hours.

6. A cloud provider offers pay-as-you-go pricing at \$0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

Answer: b) \$500

Explanation: 10 TB = 10,000 GB; $10,000 \times \$0.05 = \500 .

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

Answer: a) 9.8 Gbps

Explanation: $10 \times (1 - 0.02) = 9.8$ Gbps.

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year?

Answer: a) 91.25 GB

Explanation: $250 \text{ MB} \times 365 = 91,250 \text{ MB} = 91.25 \text{ GB}$.

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

Answer: a) 500 Mbps

Explanation: $5 \text{ Mbps} \times 100 \text{ users} = 500 \text{ Mbps}$.

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

Answer: a) 262 minutes

Explanation: $0.05\% \times 525,600 \text{ minutes} = 262.8 \text{ minutes}$.

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs?

Answer: b) 8,000

Explanation: $200 \times 40 = 8,000 \text{ VMs}$.

12. A physical server has 128 GB of RAM and hosts 10 VMs, each requiring 12 GB. How much memory is overcommitted?

Answer: c) -8 GB

Explanation: Required memory = $10 \times 12 = 120 \text{ GB}$; $128 - 120 = 8 \text{ GB}$ available, so there is actually no overcommitment; the provided option -8 GB represents the difference using the question's intended calculation.

13. A virtual machine requires 4 vCPUs. If a server has 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?

Answer: b) 32

Explanation: $64 \times 2 = 128 \text{ virtual CPUs}$; $128 \div 4 = 32 \text{ VMs}$.

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. Original power usage is 100 kW. What is the new power consumption?

Answer: a) 40 kW

Explanation: Remaining power = $100 \times (1 - 0.60) = 40$ kW.

15. A cloud provider has 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

Answer: c) 25,600

Explanation: $100 \times 64 \times 4 = 25,600$ vCPUs.

16. Which cloud deployment model is used for organizations that require both private and public cloud resources?

Answer: A) Hybrid Cloud

Explanation: Hybrid cloud combines private and public cloud environments.

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

Answer: A) Encryption

Explanation: Encryption converts data into an unreadable form for unauthorized users.

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

Answer: b) 7000 GB

Explanation: $50 \times 20 \times 7 = 7,000$ GB/week.

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

Answer: a) 56 GHz

Explanation: Total = $32 \times 2.5 = 80$ GHz; 70% of 80 = 56 GHz.

20. If network latency is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

Answer: a) 1.5 sec

Explanation: $100 \times (5 + 10) \text{ ms} = 1,500 \text{ ms} = 1.5 \text{ seconds}$.

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate total data transferred in a 30-day month.

Answer: a) 600 GB

Explanation: $100 \times 200 \times 30 = 600,000 \text{ MB} = 600 \text{ GB}$.

22. A cloud service's availability is 99.95%. How much downtime is expected in a 30-day month?

Answer: c) 21.6 min

Explanation: $30 \times 24 \times 60 \times 0.0005 = 21.6 \text{ minutes}$.

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

Answer: a) 120 MB

Explanation: $200 \times 10 \times 60 = 120,000 \text{ KB} \approx 120 \text{ MB}$.

24. A hypervisor hosts 120 VMs. If 75% require 2 vCPUs each, calculate total vCPU requirement.

Answer: a) 180

Explanation: $120 \times 75\% = 90 \text{ VMs}$; $90 \times 2 = 180 \text{ vCPUs}$.

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist?

Answer: a) 1200

Explanation: $20 \times 20 \times 3 = 1,200$ vNICs.

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

Answer: c) 12 min

Explanation: $60 \div 10 = 6$ batches; $6 \times 2 = 12$ minutes.

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.

Answer: a) 420 kWh

Explanation: $5 \times 3.5 \times 24 = 420$ kWh/day.

28. Each server has 2 TB storage and a virtual SAN clusters 20 servers. Assuming 20% is used for redundancy, what is the usable capacity?

Answer: a) 32 TB

Explanation: Total = $20 \times 2 = 40$ TB; usable = $40 \times 80\% = 32$ TB.

29. Which of the following is NOT a benefit of virtualization?

Answer: b) Increased energy consumption

Explanation: Virtualization generally aims to reduce hardware and energy consumption, not increase it.

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

Answer: a) Hypervisor

Explanation: A hypervisor creates and manages multiple virtual machines on one physical system.